THE HOMOLOGY OF THE FIBRE OVER A POINT IS THE SAME FOR THE STABILITY PARAMETERS ZETA AND ZETA-PLUS

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Allowed standard prerequisites (editorial).

- The moduli spaces M_zeta(v,w) and M_zeta^norm(lambda_i, lsp{t}{delta} w) and the projection pi_{0,zeta} from the zeta-stable locus to the zeta = 0 locus. Set up earlier in the same section and subsection (source lines 2329-2965). Fixes the fibres whose homology the statement compares, and the product decomposition used in the proof.
- The homotopy equivalence between the fibre over the origin in M_zeta(v - v', w - C v') and its zeta-plus counterpart. Used inside the counted proof at source lines 2993-2997, attributed by the author to section 5.5 of the cited paper Na-quiver. Reduces the comparison of the two fibres to the one factor that actually depends on the stability parameter.

Lemma 0.1. *Let us fix* $\mathbf{v}$ *and consider* $\pi_{0,\zeta}^{-1}(x) \subset \mathfrak{M}_\zeta(\mathbf{v},\mathbf{w})$ *and* $\pi_{0,\zeta^+}^{-1}(x) \subset \mathfrak{M}_{\zeta^+}(\mathbf{v},\mathbf{w})$. *There is a graded vector space isomorphism* $H_*(\pi_{0,\zeta}^{-1}(x)) \cong H_*(\pi_{0,\zeta^+}^{-1}(x))$ (*which is canonical in the way explained during the proof*). *In particular,* $H_{\mathrm{top}}(\pi_{0,\zeta}^{-1}(x))$ *is isomorphic to* $V^{\mathfrak{g}}(\mathbf{w}-\mathbf{v}'-|\lambda|\delta)$ *as an* $\mathfrak{h}$*-module.*

Proof. We know that $\pi_{0,\zeta}^{-1}(x) \subset \mathfrak{M}_\zeta(\mathbf{v},\mathbf{w})$ is isomorphic to the product of $\pi_{0,\zeta}^{-1}(0) \subset \mathfrak{M}_\zeta(\mathbf{v}-\mathbf{v}', \mathbf{w}-\mathbf{c}\mathbf{v}')$ and $\pi_{0,\zeta}^{-1}(0) \subset M_\zeta^{\mathrm{norm}}(\lambda_i, {}^t\delta\mathbf{w})$ $(i=1,\dots,l)$ as in the previous subsection. The same holds for ζ^+. The latter factors are the same for our ζ

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and ζ^+, so we need to worry about the first factor. We know that the inclusion $\pi_{0,\zeta}^{-1}(0) \subset \mathfrak{M}_\zeta(\mathbf{v}-\mathbf{v}', \mathbf{w}-\mathbf{C}\mathbf{v}')$ and one for ζ^+ are homotopy equivalences [4, 5.5], and the C^∞-structure of $\mathfrak{M}_\zeta(\mathbf{v}-\mathbf{v}', \mathbf{w}-\mathbf{C}\mathbf{v}')$ is independent of the choice of generic parameter ζ. In particular, we get the first assertion. Since this is the weight space, we have the second assertion. □

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